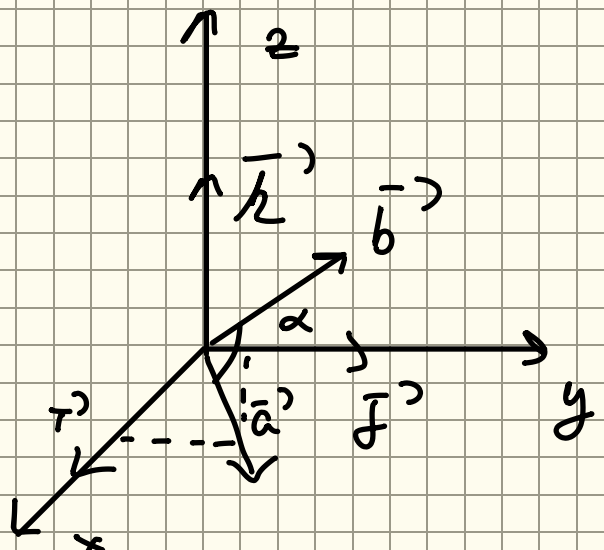


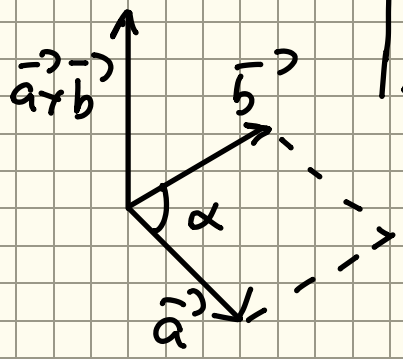
$$\Rightarrow \vec{a} = x_A\vec{i} + y_A\vec{j} + z_A\vec{k}$$

$$= (x_A, y_A, z_A)$$

$$\vec{b} = x_B\vec{i} + y_B\vec{j} + z_B\vec{k}$$

$$\vec{a}\cdot\vec{b} = x_A x_B + y_A\cdot y_B + z_A\cdot z_C = |\vec{a}|\cdot|\vec{b}|\cdot\cos\alpha$$

$$\vec{a}\times\vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_A & y_A & z_A \\ x_B & y_B & z_B \end{vmatrix} = \vec{i}\cdot\begin{vmatrix} y_A & z_A \\ y_B & z_B \end{vmatrix} - \vec{j}\cdot\ldots$$



$$\Rightarrow |\vec{a}\times\vec{b}| = |\vec{a}|\cdot|\vec{b}|\cdot\sin\alpha$$

1.1) Decompose $\vec{v} = \vec{i} + 3\vec{j} - 5\vec{k}$ as a linear combination of $\vec{a} = \vec{i} + 2\vec{j}$, $\vec{b} = \vec{i} + 2\vec{k}$, $\vec{c} = 2\vec{i} - \vec{j} + \vec{k}$

We look for scalars: $\alpha, \beta, \gamma \in \mathbb{R}$ s.t. $\vec{v} = \alpha\vec{a} + \beta\vec{b} + \gamma\vec{k} = \alpha\vec{i} + 2\alpha\vec{j} + \beta\vec{i} + 2\beta\vec{k} + 2\gamma\vec{i} - \gamma\vec{j} + \gamma\vec{k}$

$$\vec{v} = \vec{i}(\alpha+\beta+2\gamma) + \vec{j}(2\alpha-\gamma) + \vec{k}(2\beta+\gamma)$$

$$= \vec{i} + 3\vec{j} - 5\vec{k}$$

$$\Rightarrow \begin{cases} \alpha+\beta+2\gamma = 1 \\ 2\alpha - \gamma = 3 \\ 2\beta + \gamma = -5 \end{cases}$$

$$2(\alpha+\beta) = -2$$

$$\Rightarrow \alpha+\beta = -1$$

$$\Rightarrow \gamma = 1$$

$$\Rightarrow \alpha = 2$$

$$\beta = -3$$

$\Rightarrow \vec{v} = 2\vec{a} - 3\vec{b} + \vec{c}$

$\Leftrightarrow [A|B] = \begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 2 & 0 & -1 & | & 3 \\ 0 & -2 & 1 & | & -5 \end{bmatrix} \overset{L_2 - 2L_1}{\sim}$

$\begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & -2 & -5 & | & 1 \\ 0 & -2 & 1 & | & -5 \end{bmatrix} \overset{L_3 + L_2}{\sim} \begin{bmatrix} 1 & 1 & 2 & | & 1 \\ 0 & -2 & -5 & | & 1 \\ 0 & 0 & -4 & | & -4 \end{bmatrix} \Leftrightarrow \begin{cases} \alpha + \beta + 2\gamma = 1 \\ -2\beta - 5\gamma = 1 \\ -4\gamma = -4 \end{cases}$

$\Rightarrow \gamma = 1$

$\beta = -3$

$\alpha = 2$

$\Rightarrow (\alpha, \beta, \gamma) = (2, -3, 1)$

1.2) Let $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$, $\vec{b} = \vec{j} - 2\vec{k}$, $\vec{c} = \vec{i} - \vec{j} + 2\vec{k}$

Compute:

a) $\vec{a} + \vec{b} = ?$

b) $|\vec{b}| = ?$

c) $\vec{b} \cdot \vec{c}$

d) $\vec{a} \times \vec{b}$

e) $(\widehat{\vec{a}, \vec{b}}) = ?$

f) $\vec{a} \cdot (\vec{b} + \vec{c})$

g) $(\vec{a} \times \vec{b}) \cdot \vec{c}$

h) The projection of $\vec{a}$ onto the vector $\vec{b}$ $(\vec{b}, \mathrm{pr}_{\vec{b}}, \vec{a})$

a) $\vec{a} + \vec{b} = 2\vec{i} - 2\vec{j} - \vec{k}$

b) $|\vec{b}| = \sqrt{1^2 + 2^2} = \sqrt{5}$

c) $\vec{b} \cdot \vec{c} = |\vec{b}| \cdot |\vec{c}| \cdot \cos 0 = -1 - 4 = -5$

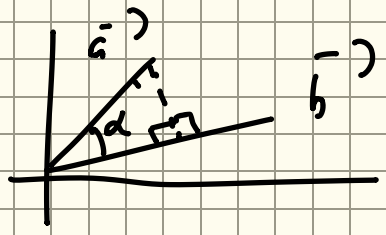
d) $\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 1 \\ 0 & 1 & -2 \end{vmatrix} = \vec{i} \begin{vmatrix} -3 & 1 \\ 1 & -2 \end{vmatrix} - \vec{j} \begin{vmatrix} 2 & 1 \\ 0 & -2 \end{vmatrix} + \vec{k} \begin{vmatrix} 2 & -3 \\ 0 & 1 \end{vmatrix} = 5\vec{i} + 4\vec{j} + 2\vec{k}$

e) $\cos\theta = \frac{-3-2}{\sqrt{14}\cdot\sqrt{5}} = \frac{-5}{\sqrt{70}} \Rightarrow \theta = \arccos\left(\frac{-5}{\sqrt{70}}\right)$

f) $\vec{a}\cdot(\vec{b}+\vec{c}) = 2\cdot 1 = 2$

g) $(\vec{c}\times\vec{b})\cdot\vec{c} = 5-4+4 = 5$

h)



$$pr_{\vec{b}}\vec{c} = |\vec{a}|\cdot|\cos\alpha| = |\sqrt{14}|\cdot\left|\frac{-5}{\sqrt{70}}\right| = \sqrt{5}$$

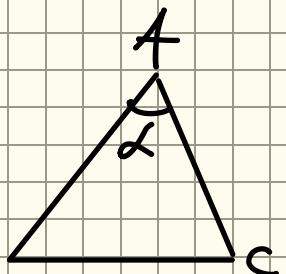
1.3) Consider 3 points $A(-2,1,0)$, $B(2,0,3)$, $C(-2,0,4)$, $D(-4,1,3)$. Determine:

a) $A_{\Delta ABC}$ d) V_{ABCD}

b) $d(D, BC)$ e) h_{ABCD}, with base ABC

c) $\measuredangle(\vec{AB}, \vec{CD})$

a)



$$A_{ABC} = \frac{1}{2}\cdot|\vec{AB}|\cdot|\vec{AC}|\cdot\sin\alpha$$
$$= \frac{1}{2}\cdot|\vec{AB}\times\vec{AC}|$$

$\vec{AB} = 4\vec{i} - \vec{j} + 3\vec{k}$

$\vec{AC} = -\vec{j} + 4\vec{k}$

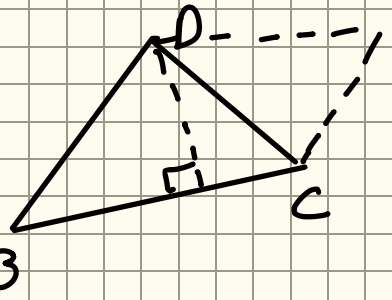
$\vec{AB}\times\vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & -1 & 3 \\ 0 & -1 & 4 \end{vmatrix}$

$= \vec{i}\cdot\begin{vmatrix} -1 & 3 \\ -1 & 4 \end{vmatrix} - \vec{j}\begin{vmatrix} 4 & 3 \\ 0 & 4 \end{vmatrix} + \vec{k}\begin{vmatrix} 4 & -1 \\ 0 & -1 \end{vmatrix} = -\vec{i} - 16\vec{j} - 4\vec{k}$

$|\vec{AB} \times \vec{AC}| = \sqrt{1 + 16^2 + 4^2} = \sqrt{17 + 256} = \sqrt{273}$

$A_{ABC} = \frac{\sqrt{273}}{2}$

b)



$A_{D,BC} = |\vec{BD} \times \vec{BC}|$

$A_{D,BC} = d(D,BC) \cdot |\vec{BC}|$

$\Rightarrow d(D,BC) = \frac{|\vec{BD} \times \vec{BC}|}{|\vec{BC}|}$, $\vec{BD} = -6\vec{i} + \vec{j}$

$\vec{BC} = -4\vec{i} + \vec{k}$

$$\vec{BD} \times \vec{BC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -6 & 1 & 0 \\ -4 & 0 & 1 \end{vmatrix} = \vec{i} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \vec{j} \begin{vmatrix} -6 & 0 \\ -4 & 1 \end{vmatrix} + \vec{k} \begin{vmatrix} -6 & 1 \\ -4 & 0 \end{vmatrix}$$

$= \vec{i} + 6\vec{j} + 4\vec{k}$

$d(D,BC) = \frac{\sqrt{1+36+16}}{\sqrt{16+1}} = \frac{\sqrt{53}}{\sqrt{17}}$

c) $\cos\alpha = \frac{\vec{AB} \cdot \vec{CD}}{|\vec{AB}| \cdot |\vec{CD}|}$

$|\vec{AB}| = \sqrt{16+1+9} = \sqrt{26}$

$|\vec{CD}| = \sqrt{36+1+1} = \sqrt{38}$

$\vec{AB} \cdot \vec{CD} = -24 - 1 - 9 = -34 \Rightarrow \alpha = \arccos\left(\frac{-34}{\sqrt{26} \cdot \sqrt{38}}\right)$

d) $V_{ABCD} = \frac{1}{6} \cdot |(\vec{AB}, \vec{AC}, \vec{AD})|$. ABCD piramida

$\vec{AB} = (4, -1, 3)$
$\vec{AC} = (0, -1, 4)$
$\vec{AD} = (-2, 0, 3)$

$\Rightarrow \begin{vmatrix} 4 & -1 & 3 \\ 0 & -1 & 4 \\ -2 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 0 & -1 & 9 \\ 0 & -1 & 4 \\ -2 & 0 & 3 \end{vmatrix}$

$= -2 \cdot (-1)^4 \cdot \begin{vmatrix} -1 & 9 \\ -1 & 4 \end{vmatrix} = -2(-4+9) = -10$

$V_{ABCD} = \frac{1}{6} \cdot |-10| = \frac{5}{3}$

e) $V_{ABCD} = \frac{d(D, ABC) \cdot A_{ABC}}{3} \Rightarrow d(D, ABC) = \frac{3 V_{ABCD}}{A_{ABC}}$

$= \frac{3 \cdot \frac{5}{3}}{\frac{\sqrt{27}}{2}} = \frac{10}{\sqrt{27}}$

1.5) Determine $\lambda \in \mathbb{R}$ s.t. $\vec{a} = \vec{i} + 2\lambda\vec{j} - (\lambda - 1)\vec{k}$, $\vec{b} = (3 - \lambda)\vec{i} + \vec{j} + 2\vec{k}$: $a \perp b$

$\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0 \Rightarrow 3 - \lambda + 2\lambda - 2\lambda + 2 = 0$

$\Rightarrow \lambda = 5$

1.6) Determine the angle between $\vec{u}$ and $\vec{v}$, if: $|\vec{v}| = 2$, $|\vec{v}| = 4$ and $2\vec{u} + \vec{v} \perp (3\vec{u} - 2\vec{v})$

$(2\vec{u} + \vec{v}) \cdot (3\vec{v} - 2\vec{v}) = 0$

$6|\vec{u}|^2 - 4\vec{u} \cdot \vec{v} + 3\vec{v} \cdot \vec{v} - 2|\vec{v}|^2 = 0$

$6 \cdot 4 - \vec{u} \cdot \vec{v} - 2 \cdot 16 = 0 \iff \vec{u} \cdot \vec{v} = -8$

$$\Rightarrow \quad \cos\alpha = \frac{-8}{2\cdot 4} = -1 \Rightarrow \alpha = \pi$$